

Provably Correct Post-Update Incremental Maintenance of RDB2RDF Views

Vânia Maria Ponte Vidal
Federal University of Ceará
Fortaleza, Brazil
vvidal@dc.ufc.br

José Renato da Silva Freitas
Federal University of Ceará
Fortaleza, Brazil
jrenatosfreitas@gmail.com

Narciso Arruda
Federal University of Ceará
Fortaleza, Brazil
narcisoarruda@gmail.com

Marco A. Casanova
Tecgraf Institute and Department of Informatics, PUC-Rio
Rio de Janeiro, Brazil
casanova@inf.puc-rio.br

Abstract

A strategy for exposing relational data as part of a knowledge graph is to define RDB2RDF views that map relational tuples to RDF triples using transformation rules. These views are often materialized in RDF stores to improve query performance and data availability, a process that requires incremental mechanisms to maintain them during updates. A key problem in incremental maintenance is computing the changesets to apply to a view, which traditionally depend on the pre-update state of the database. The central contribution of this paper is a formal model of transformation rules that yields provably correct changesets operating directly on the post-update state. The second contribution is to propose an operational architecture for incremental maintenance of RDB2RDF views that realizes the formal model. The architecture presupposes that the view is defined by R2RML statements and features AFTER triggers and a maintenance worker to implement transformation rules generated from the R2RML statements. The third contribution is a feasibility study of the operational architecture based on an RDB2RDF view, defined by R2RML statements, on a version of the MusicBrainz database.

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PVLDB Artifact Availability:

The source code, data, and/or other artifacts have been made available at <https://github.com/semantic-ekgraphs/IVM>.

1 Introduction

Enterprise Knowledge Graphs (EKGs) have emerged as a powerful paradigm for integrating heterogeneous enterprise data sources into a unified semantic layer. By linking structured and semi-structured data via an ontology, EKGs enable applications to query and analyze

enterprise information in a semantically rich, integrated manner. In many real-world deployments, the underlying enterprise data remains stored in relational databases, while the knowledge graph provides a semantic access layer on top of these sources.

A common strategy for exposing relational data as part of a knowledge graph is the definition of *RDB2RDF views*, which map relational tuples into RDF triples using transformation rules. These views are often materialized in RDF stores to improve query performance and data availability. However, when the relational source is frequently updated, the materialized RDB2RDF view must be continuously synchronized with the underlying database. Recomputing the entire RDB2RDF view after each update is typically impractical, motivating the need for *incremental maintenance techniques* that compute only the changes to the view.

A well-established approach for incremental view maintenance is to compute *changesets*, represented as a pair $\langle \Delta^-(u), \Delta^+(u) \rangle$, capturing the triples removed from and added to the RDB2RDF view after a relational update u . Prior work [19] has shown that, for entity-preserving RDB2RDF statements, it is sufficient to identify the *pivot tuples* whose RDF contributions may be affected by the update and recompute their associated RDF state.

The formal framework introduced in [19] provides a rigorous method for computing such changesets based on transformation rules over relational schemas. In particular, in this framework, the deletion component $\Delta^-(u)$ is defined with respect to the pre-update database state. Conceptually, this definition is correct, as it captures precisely the RDF triples that must be removed from the view. However, it introduces a practical limitation when implementing the framework with database triggers, namely, relying on the pre-update database state.

The central contribution of this paper is to challenge this assumption by introducing a formal model that yields provably correct changesets operating directly on the post-update state. It shows that, for a transformation rule without multiple occurrences of the same relation, both deletion and insertion components can be derived using only the post-update state, the transition tables, and the pre-update view state. For a transformation rule with multiple occurrences of the same relation R , it captures the impact of deletions and insertions on R directly in the evaluation of the transformation rule, still without using the pre-update database state.

The second contribution of the paper is to propose an operational architecture for incremental maintenance of RDB2RDF views

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that realizes the formal model. The architecture decouples SPARQL computation from relational transactions, thereby avoiding the execution of SPARQL queries inside relational transactions and isolating the triplestore. The architecture presupposes that the RDB2RDF view is defined by R2RML statements, which requires a preprocessing step to translate such statements into transformation rules. The paper proposes implementing this preprocessing step using an LLM prompted with the formal model. Then, the paper discusses how to translate the transformation rules into AFTER triggers, again using an LLM prompted with the formal model. Taken together, this means that, given an RDB2RDF view defined by R2RML statements, the underlying relational database can be equipped with AFTER triggers that, combined with the other (fixed) components of the architecture, incrementally maintain the RDB2RDF view.

The third contribution of the paper is a feasibility study of the operational architecture. The study evaluates the complete implementation cycle for an RDB2RDF view built on a version of the MusicBrainz database. The database features a schema with 374 relations that encompass information about artists, release groups, etc. The RDB2RDF view uses the *Music Ontology* vocabulary and is defined by 57 R2RML statements.

To summarize, existing incremental RDB2RDF maintenance formulations are difficult to realize using statement-level AFTER triggers because their deletion changesets depend on the pre-update relational state, which is no longer directly available when such triggers execute. The proposed formal model removes this mismatch by reformulating the computation of changesets in terms of the post-update relational state, the transition tables, and the pre-update RDF view. Equally important, the resulting changesets are provably correct. The formal model is also sufficiently constructive to support specification-guided generation of executable maintenance triggers from R2RML mappings.

Lastly, it should be stressed that an LLM is not used to establish the correctness of the changesets. Rather, the LLM-based study provides independent operational evidence that the formal model is sufficiently constructive to be translated into executable maintenance code.

The paper is organized as follows. Section 2 covers related work. Section 3 introduces the notion of entity-preserving RDB2RDF views. Section 4 summarizes the theory of incremental RDB2RDF view maintenance. Section 5 discusses the operational architecture. Section 6 describes the experiments conducted and discusses the results. Finally, Section 7 contains the conclusions.

2 Related Work

Incremental View Maintenance (IVM). The Incremental View Maintenance problem has been studied in the literature for a very long time [11]. Early work covered relational views [5], object-oriented views [2], semi-structured views [24], and XML views [8], just to name a few efforts. Some more recent approaches include the following. As for generic IVM tools, DBSP [4] introduces an expressive language that enables efficient incremental view maintenance for queries written in multiple languages, including SQL and Datalog. Enzyme [23] is an IVM engine developed at Databricks to power Spark Declarative Pipelines, providing a built-in, end-to-end approach to incremental pipelines that utilizes materialized

views as first-class building blocks. OpenIVM [3] is an SQL-to-SQL compiler for Incremental View Maintenance (IVM) that compiles view definitions into SQL to propagate deltas into the table that materializes the view eventually. `pg_ivm`¹ is the most prominent, production-ready implementation of IVM for PostgreSQL.

As for IVM SPARQL and graph databases, Incremunica [17] integrates multiple incremental operators, allowing it to adapt to different queries and data for optimal performance, and supporting federated querying, dynamically adding and removing sources during query execution, SPARQL Query Language support, multiple IVM techniques, and client-side execution. The counting algorithm, a classic approach to incremental view maintenance for relational queries, is adapted in [15] to SPARQL queries over RDF datasets. Graph property views, both virtual and materialized, are considered in [6] for integrating heterogeneous data using rewriting techniques.

Solutions in specific settings include the following. The maintenance of self-join-free conjunctive queries over a dynamic database, where tuples can be inserted or deleted, is investigated in [7]. A maintenance approach that combines delta queries, trees of materialized views, and heavy-light data partitioning is introduced in [1], whose update time matches or improves the best update time reported in prior work. The incorporation of knowledge about foreign keys into IVM to speed up performance is investigated in [16], which reports experiments in Redshift showing that the proposed technique improved the execution times of the whole refresh process up to 2 times, and up to 2.7 times the process of calculating the necessary changes that will be applied to the materialized view. Finally, a framework for maintaining sketches incrementally under updates is introduced in [9].

The main idea that distinguishes the formal model proposed in [20] from previous work on incremental view maintenance is to explore the entity-preservation property of typical RDB2RDF views, which enables the identification of the specific tuples relevant to a data source update. Only the portion of the view associated with the relevant tuples should be rematerialized. This earlier work was further modified and enhanced to compute changesets for RDB2RDF views [19].

However, such earlier approaches depended on accessing the database state before the update to generate the view changesets. The formal model proposed in this paper addresses this situation by computing the view changesets from the post-update database state, the current view state, and the update itself.

The definition of the formal model proposed in this paper co-evolved with the prompts used in Section 6. An LLM served as an operationalization instrument, and not to establish the correctness of the proposed model, which follows from the formal results. During the development of the formal model, attempts to generate executable maintenance logic from the definitions exposed implicit assumptions and formulations that, although meaningful, were not sufficiently constructive for direct implementation. The resulting feedback led to clearer definitions and a more explicit connection between the formal changeset semantics and the data available to statement-level AFTER triggers.

¹https://github.com/sraoss/pg_ivm

Specification-guided LLM synthesis. Specification-guided LLM synthesis is the process of generating executable artifacts from formally defined semantic specifications, in which the generative model is constrained to preserve the specification’s structural and logical properties.

For example, Murphy et al. [10] divided code generation into two parts: one handled by an LLM and the other by formal methods-based program synthesis. VERGE [14] decomposes LLM outputs into atomic claims, autoformalizes them into first-order logic, and verifies their logical consistency using automated theorem proving. SGCR [22] is a framework that grounds LLMs in human-authored specifications to produce trustworthy and relevant feedback.

Specifically with respect to SQL code generation, the text-to-SQL task, that is, the generation of SQL from Natural Language descriptions, has received considerable attention. Recent LLM-based text-to-SQL tools are surveyed, for example, in [12, 13]. The Text-to-SQL Handbook² lists the best-performing text-to-SQL tools on several benchmarks.

This paper takes a different direction and shows that LLMs are capable of processing a (complex and lengthy) formal model and outputting the formally specified database code. Specifically, the paper first shows how to translate R2RML statements into transformation rules using an LLM prompted with the formal model. It then demonstrates how to translate the transformation rules into the required AFTER triggers, again using an LLM prompted with the formal model.

3 Formal Specification of Entity-Preserving RDB2RDF Views

This section summarizes the major concepts and results of a theory of incremental RDB2RDF view maintenance, presented in detail in the companion extended paper [18].

3.1 Entity-Preserving RDB2RDF Views

An *RDB2RDF view* is a triple $\mathcal{W} = (V, \mathcal{S}, M)$, where:

- V is the vocabulary of the *target ontology*;
- $\mathcal{S}$ is the *source relational schema*; and
- M is a set of *mappings* between V and $\mathcal{S}$, defined by *transformation rules* (TRs).

The semantics of an RDB2RDF view is defined in Section 3.3. A view $\mathcal{W} = (V, \mathcal{S}, M)$ satisfies the *entity-preserving property* iff:

- the instances of the classes in V correspond to tuples of selected relations of $\mathcal{S}$, so that distinct tuples from the same relation generate distinct class instances. These tuples and relations are called *pivot tuples* and *pivot relations*;
- the values of datatype properties in V of these instances are given by (functions of) attribute values of the corresponding tuples, or of related tuples;
- the object properties in V correspond to relationships between tuples of the pivot relations of $\mathcal{S}$.

A *transformation rule* (TR) of $\mathcal{W}$ is an expression of the form $C(x) \leftarrow Q(x)$ or of the form $P(x,y) \leftarrow Q(x,y)$, where C and P are class and property names in V , and $Q(x)$ and $Q(x,y)$ are queries over $\mathcal{S}$ whose target clauses contain one and two variables, respectively.

²https://github.com/HKUSTDial/NL2SQL_Handbook

The problem of generating TRs is addressed in [20, 21] and is outside of the scope of this work. Section 3.2 contains examples of TRs for the RDB2RDF view used in the experiments.

Table 1 lists a set of TR patterns that lead to the definition of relational to RDF statements that guarantee that the RDB2RDF view satisfies the entity-preserving property by construction. Table 1 defines three types of TR patterns: *Class Transformation Rule* (CTR), *Datatype Property Transformation Rule* (DTR), and *Object Property Transformation Rule* (OTR).

Intuitively, a CTR Ψ maps tuples of a relation R into RDF instances of a class C . It establishes a semantic correspondence between a pivot tuple r and an RDF resource x , such that each pivot tuple is associated with at most one RDF instance, and distinct pivot tuples generate distinct RDF resources. Thus, the mapping preserves the identity of relational entities in the RDB2RDF view.

There are two types of DTRs. A *Local DTR* has no relational path in the body so that the datatype values are obtained directly from attribute values of the pivot tuples.

A *Path DTR* has a relational path in the body, and the datatype values are obtained from tuples reachable through the relational path from the pivot tuples.

There are also two types of OTRs. A *regular OTR* Ψ defines object properties by relating an RDF instance generated from a pivot tuple t to an RDF instance generated from a pivot tuple u reached from t through a relational path in the source schema. Hence, OTRs capture relationships between entities represented in the database.

A *reification OTR* relates an RDF instance generated from a pivot tuple t to a URI³ obtained by reification of a literal l created from attribute values of a pivot tuple u reached from t through a relational path in the source schema.

The transformation rules in Table 1 and in the experiments use several predicates. The simplest ones are:

- $nonNull(v)$ holds iff value v is not null.
- $RDFLiteral(u, A, R, v)$ holds iff v is the literal representation of a value u of attribute A of R .
- $RDFIRI(u, o)$ holds iff the non-null value u is converted into the RDF IRI o (that is, o is just the literal u treated as a URI).
- $F(r, s)$ holds iff a tuple r of R is related to tuple s of S by a foreign key of the form $F(R:L, S:K)$.

The more complex predicates are as follows.

A *relational path* φ is a list of the form “ $[F_1, \dots, F_n]$ ”, where F_j is either the name of a foreign key, indicating that the path traverses F_j from R_j and R_{j+1} , or of the form $(F_j)^{-1}$, indicating that the path traverses F_j from R_{j+1} to R_j , in the reverse direction. The expression $path(\Psi)$ denotes the relational path of Ψ . A relational path φ induces a *predicate path* $\varphi(x, y)$, which is evaluated in a database state by joining the relations in the path in the directions indicated.

Let R be the pivot relation of a transformation rule Ψ , and r be a pivot tuple in R .

A *URI template* for R is a string pattern p with place holders $\langle h_1, \dots, h_n \rangle$ such that, when the place holders are appropriately instantiated, the resulting string becomes a URI.

³Throughout this paper, the term URI is used for consistency with the URI-constructor notation. Formally, every URI occurring in an RDF triple denotes an RDF IRI.

Table 1: Transformation Rule Patterns

TR	Transformation Rule Pattern
CTR	$\psi : C(s) \leftarrow R(r), URI_R(r, s), \delta[r]$ where - ψ is the name of the transformation rule - R is the name of a relation or a view in the source schema S - r is a tuple variable bound to tuples of R - s is the URI associated with the tuple bound to r - $URI_R(r, s)$ is a URI constructor predicate as defined in the text - $\delta[r]$ is an optional selection over R , among other relations
Local DTR	$\psi : P(s, v) \leftarrow R_D(r), B_D[r, s], nonNull(r.A_1), \dots, nonNull(r.A_k),$ $RDFLiteral(r.A_1, "A'_1", "G'', v_1), \dots, RDFLiteral(r.A_k, "A'_k", "G'', v_k), T([v_1, \dots, v_k], v), \delta[r]$ where - ψ is the name of the transformation rule - P is a datatype property of V with domain D - $nonNull$ and $RDFLiteral$ are predicates as defined in the text - " $R_D(r), B_D[r, s]$ " is the RHS of the CTR ψ_D that matches class D with pivot relation R_D of ψ_D - $A_1, \dots, A_k$ are attributes of R_D - T is an optional function that transforms values of attributes $A_1, \dots, A_k$ of R_D to values of property P - $\delta[r]$ is an optional selection over R_D , among other relations
Path DTR	$\psi : P(s, v) \leftarrow R_D(r), B_D[r, s], nonNull(r.A_1), \dots, nonNull(r.A_k),$ $RDFLiteral(r.A_1, "A'_1", "G'', v_1), \dots, RDFLiteral(r.A_k, "A'_k", "G'', v_k), T([v_1, \dots, v_k], v),$ $\varphi(r, g), \delta[r, g]$ where - $\varphi(r, g)$ is a predicate path starting on R_D and ending on a relation R_G - $\delta[d, g]$ is an optional selection over R_D and R_G , among other relations - the other terms are as in a Path DTR
Regular OTR	$\psi : P(s, o) \leftarrow R_D(d), B_D[d, s], R_G(g), B_G[g, o], \varphi(d, g), \delta[d, g]$ where - " $R_G(g), B_G[g, o]$ " is the RHS of the CTR ψ_G that matches class G with pivot relation R_G of ψ_G - the other terms are as in a Path DTR
Reification OTR	$\psi : P(s, o) \leftarrow R_D(d), B_D[d, s], nonNull(r.A_1), \dots, nonNull(r.A_k),$ $RDFLiteral(r.A_1, "A'_1", "G'', v_1), \dots, RDFLiteral(r.A_k, "A'_k", "G'', v_k), T([v_1, \dots, v_k], v),$ $RDFIRI(u, o), \delta[r]$ where - $RDFIRI(u, o)$ is as defined in the text - the other terms are as in a Local DTR

A URI s is associated with r via p iff the predicate

$$hasURI(p, \langle e_1, \dots, e_n \rangle, s)$$

holds, where

- $\langle e_1, \dots, e_n \rangle$ is an ordered sequence of expressions, where each e_i is of the form:
 - $e_i = r.A$, where $r.A$ denotes an attribute of the pivot tuple.
 - $e_i = [FK(r, t)/t.A]$, where $[FK(r, t)/t.A]$ denotes the value of attribute A obtained after following the foreign key FK from the pivot tuple r to tuple t such that the foreign-key navigation must be functional from the pivot relation to the referenced relation.
- s is the result of instantiating each h_i by e_i in p .

Finally, to simplify the notation, a second predicate, called a *URI constructor*, is introduced by definition:

$$URI_R(r, s) \equiv hasURI(p, \langle e_1, \dots, e_n \rangle, s)$$

3.2 Examples of Transformation Rules

The case study of Section 6 uses an RDB2RML view defined on top of the *MusicBrainz* database. The view adopts the *Music Ontology* vocabulary and is defined using R2RML statements, which are automatically translated into transformation rules (see Section 5.3).

Figure 1(a) shows the *MusicBrainz* schema, Figure 1(b) illustrates a fragment of the *Music Ontology*, and Table 2 lists some of the transformation rules compiled from the R2RML statements.

Table 2 is organized as follows:

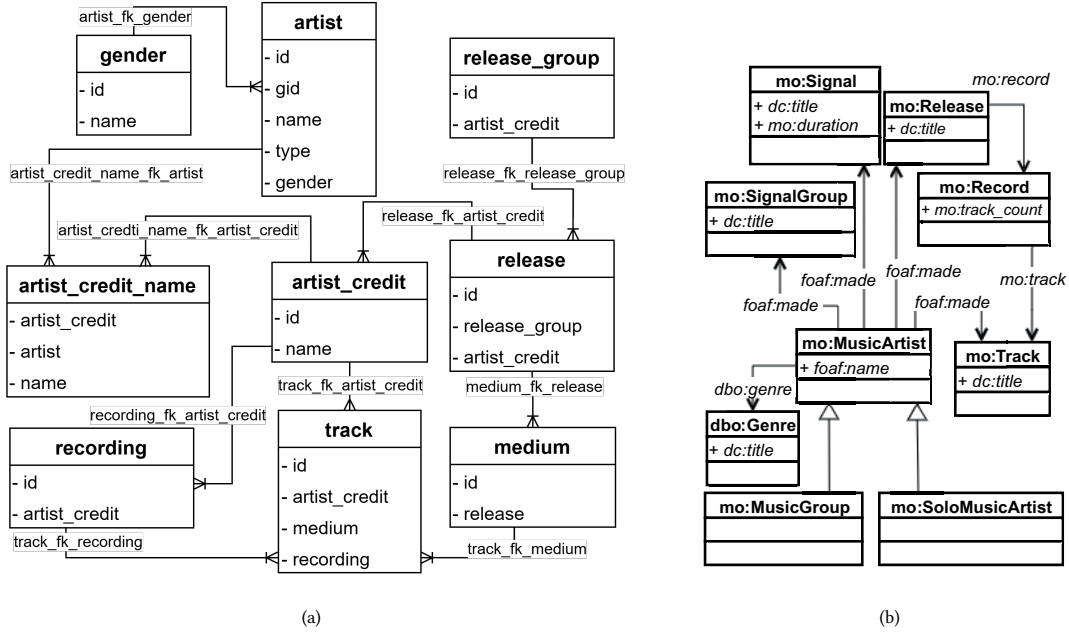


Figure 1: (a) *MusicBrainz Schema* and (b) *Fragment of LMB View Ontology*.

Table 2: Examples of transformation rules.

TR	Transformation Rules
	$URI_artist(r, s) \equiv hasURI("http://musicbrainz.org/artist/\langle h \rangle", \langle r.gid \rangle, s)$
	$URI_artist_tag(r, s) \equiv hasURI("http://musicbrainz.org/tagging/\langle h_1, h_2 \rangle",$ $\langle [artist_tag_fk_artist(r, a)/a.gid], [artist_tag_fk_tag(r, t)/t.name] \rangle, s)$
	$URI_medium(r, s) \equiv hasURI("http://musicbrainz.org/record/\langle h \rangle", \langle r.id \rangle, s)$
	$URI_track(r, s) \equiv hasURI("http://musicbrainz.org/track/\langle h \rangle", \langle r.id \rangle, s)$
Ψ_1	$mo:MusicArtist(s) \leftarrow artist(r), URI_artist(r, s)$
Ψ_2	$mo:SoloMusicArtist(s) \leftarrow artist(r), URI_artist(r, s), (r.type = 1)$
Ψ_3	$mo:MusicGroup(s) \leftarrow artist(r), URI_artist(r, s), (r.type = 2)$
Ψ_4	$mo:Record(s) \leftarrow medium(r), URI_medium(r, s)$
Ψ_5	$mo:Track(s) \leftarrow track(r), URI_track(r, s)$
Ψ_6	$foaf:made(s, g) \leftarrow artist(r), URI_artist(r, s), track(t), URI_track(t, g),$ $[(artist_credit_name_fk_artist)^{-1}, artist_credit_name_fk_artist_credit, (track_fk_artist_credit)^{-1}](r, t)$
Ψ_7	$mo:track(s, g) \leftarrow medium(r), URI_medium(r, s), track(t), URI_track(t, g), [(track_fk_medium)^{-1}](r, t)$
Ψ_8	$foaf:name(s, v) \leftarrow artist(r), URI_artist(r, s), nonNull(r.name), RDFLiteral(r.name, "name", "artist", v)$
Ψ_9	$foaf:gender(s, v) \leftarrow artist(r), URI_artist(r, s), gender(g), nonNull(g.name), lower(g.name, u)$ $RDFLiteral(u, "name", "gender", v), [artist_fk_gender](r, g),$

URI definitions. The first four rows illustrate how URIs are constructed. For example, the first row indicates that the URIs associated with the pivot tuples of relation *artist* are defined by

$$hasURI("http://musicbrainz.org/artist/\langle h \rangle", \langle r.gid \rangle, s)$$

That is, the URIs associated with the pivot tuple *r* is generated by replacing *h* by "*r.gid*" in the template "*http://musicbrainz.org/artist/\langle h \rangle*".

Class Transformation Rules (CTRs). Ψ_1 through Ψ_5 are CTRs that define the instances of classes *mo:MusicArtist* through *mo:Track*.

Object Property Transformation Rules (OTRs). Ψ_6 and Ψ_7 are OTRs. The body of Ψ_6 has a predicate path

$$[(\text{artist_credit_name_fk_artist})^{-1}, \\ \text{artist_credit_name_fk_artist_credit}, \\ (\text{track_fk_artist_credit})^{-1}](r, t)$$

composed of three foreign keys (defined in Figure 1(a)) that connects tuples r of relation *artist* with tuples t of relation *track*. Note that the first and the third foreign keys should be traversed in the reverse direction.

The second OTR, Ψ_7 , has a predicate path

$$[(\text{track_fk_medium})^{-1}](r, t)$$

composed of a single foreign key (again defined in Figure 1(a)) that connects tuples r of relation *medium* with tuples t of relation *track*. Again, the foreign key should be traversed in the reverse direction.

Datatype Property Transformation Rules (DTRs). Ψ_8 is a Local DTR that defines the datatype property *foaf:name* as the name of an artist, whereas Ψ_9 is a Path DTR that defines the datatype property *foaf:gender* as the artistic gender of an artist, using table *gender*.

3.3 Hierarchical RDF-State Semantics of RDB2RDF Views

Let $\mathcal{W} = (V, \mathcal{S}, M)$ be an RDB2RDF view and σ be a database state.

The *state* of $\mathcal{W}$ in σ is the set of quads (s, p, o, g) generated by the TRs in M when applied to σ , where the fourth element g is the URI of the TR that generated the triple (s, p, o) . Intuitively, each TR Ψ induces a named graph containing exactly the triples generated by Ψ , which provides a simple and operational mechanism for provenance tracking and avoids ambiguity in the presence of duplicate triples generated by different TRs. The state of $\mathcal{W}$ is defined compositionally in three hierarchical levels.

Let $\Psi \in M$ be a TR with tuple variables $r_1, \dots, r_n$ associated with relation schemes $R_1, \dots, R_n$. For tuples $p_1 \in R_1(\sigma), \dots, p_n \in R_n(\sigma)$, the *application of Ψ under σ* is the set of RDF triples produced by evaluating Ψ with bindings $r_i = p_i$, for $i = 1, \dots, n$, over σ , denoted

$$\Psi[r_1 \leftarrow p_1, \dots, r_n \leftarrow p_n](\sigma)$$

Assume that the pivot relation of Ψ is R^* . For a pivot tuple $t \in R^*(\sigma)$, the *RDF contribution of t under Ψ and σ* is the set:

$$\text{RDF_State}[\Psi](t, \sigma) = \\ \{(s, p, o, g) \mid (s, p, o) \in \Psi[r \leftarrow t](\sigma), g = \text{URI}(\Psi)\}$$

Finally, the *RDF state of the view $\mathcal{W}$ under σ* is the set

$$\text{RDF_State}(\mathcal{W}, \sigma) = \bigcup_{\Psi \in M} \bigcup_{t \in \text{pivot}(\Psi)(\sigma)} \text{RDF_State}[\Psi](t, \sigma)$$

That is, the state of an RDB2RDF view is obtained by aggregating the RDF contributions generated by all TRs for all pivot tuples in the database state.

4 Post-Update Incremental Maintenance of RDB2RDF Views

4.1 The Incremental Maintenance Problem

An *update* on a relation R is a pair $u = (D, I)$, where D and I are (possibly empty) sets of tuples, indicating that the tuples in D must be first deleted from R and then the tuples in I must be inserted into R . The application of u induces a transition from a *pre-update* database state σ_0 to the *post-update* state σ_1 .

Let u be an update and σ_0 and σ_1 be the pre-update and post-update database states of u . Let $W_0 = \text{RDF_State}(\mathcal{W}, \sigma_0)$ and $W_1 = \text{RDF_State}(\mathcal{W}, \sigma_1)$ be the view states before and after the update u .

A *view changeset* is a pair $\langle \Delta^-(u), \Delta^+(u) \rangle$, where $\Delta^-(u)$ is the set of quads to be removed from W_0 , called the *deletion changeset*, and $\Delta^+(u)$ is the set of quads to be inserted into W_0 , after deleting the quads in $\Delta^-(u)$, called the *insertion changeset*.

The view changeset is *correct* iff

$$W_1 = (W_0 \setminus \Delta^-(u)) \cup \Delta^+(u)$$

In other words, the changeset $\langle \Delta^-(u), \Delta^+(u) \rangle$ is correct if applying it to the pre-update view state W_0 yields the same result as recomputing the view over σ_1 .

An approach for computing changesets is to use database triggers. However, in relational systems, transition tables such as *deleted_R* and *inserted_R* are only available in *AFTER statement-level triggers*, where the database is already in the post-update state σ_1 . As a consequence, the pre-update state σ_0 is no longer directly accessible. A key limitation of previous approaches is that they often require reconstructing portions of the pre-update state of the relation R being updated to compute $\Delta^-(u)$. This reconstruction can be expensive and impractical, particularly for large relations. Section 4.2 then introduces a strategy for computing $\Delta^-(u)$ and $\Delta^+(u)$ that avoids this problem.

4.2 A Post-Update Formulation of Changeset Computation

This section presents an algorithm to compute the insertion and deletion changesets $\Delta^-(u)$ and $\Delta^+(u)$. The algorithm organizes the definitions in [18] into pseudocode, following a compositional structure, and ensures that all affected RDF quads are computed using only the post-update state, the tuples recorded in the transition tables, and the pre-update RDB2RDF view state.

Let $\mathcal{W} = (V, \mathcal{S}, M)$ be an RDB2RDF view and Ψ be a transformation rule (TR) in M . Let $u = (D, I)$ be an update over a relation R , and σ_1 be the database state after applying u .

The algorithm proceeds by decomposing the computation of the changeset into TR-level contributions. The first step is to identify the TRs that are relevant to updates on R :

- (1) Ψ is *pivot-relevant* to updates on R iff $\text{pivot}(\Psi) = R$, that is, R is the first relation in the body of Ψ .
- (2) Ψ is *relation-relevant* to updates on R iff R occurs in the body of Ψ in an occurrence other than the pivot occurrence.
- (3) Ψ is *relevant* to updates on R iff Ψ is pivot-relevant or relation-relevant to updates on R ; $\text{Relev}(R)$ denotes the set of transformation rules relevant to updates on R .

Note that Ψ may be both pivot-relevant and relation-relevant to R . If Ψ is pivot-relevant to updates on R , the tuples in D and I are themselves pivot tuples of Ψ , and their RDF contributions are directly affected by the update. If Ψ is relation-relevant to updates on R , the update affects the evaluation of Ψ indirectly: tuples in the pivot relation R^* of Ψ whose derivations depend on tuples in D or I may have their RDF contributions modified.

For example, consider the TR Ψ_6 in Table 2. Then, Ψ_6 is relevant to *artist*, which is the pivot relation of Ψ_6 , to *track*, which explicitly occurs in $path(\Psi_6)$, and to *artist_credit_name* and *artist_credit*, which are used in foreign keys that occur in $path(\Psi_6)$.

Next, for each rule $\Psi \in Rel_{\Psi}(R)$, the algorithm computes the contribution of Ψ to the changesets by deriving the deletion and insertion components, denoted $\Delta^-[\Psi](u)$ and $\Delta^+[\Psi](u)$, respectively, which capture the RDF quads that must be removed from and added to the view, using Ψ , with respect to u .

The overall changeset is then obtained by aggregating the contributions of all relevant rules:

$$\Delta^-(u) = \bigcup_{\Psi \in Rel_{\Psi}(R)} \Delta^-[\Psi](u) \quad \Delta^+(u) = \bigcup_{\Psi \in Rel_{\Psi}(R)} \Delta^+[\Psi](u)$$

The key idea is that changesets can be computed by characterizing the effect of an update on each relevant TR contribution.

The deletion and insertion components, $\Delta^-[\Psi](u)$ and $\Delta^+[\Psi](u)$, in turn have two components:

$$\Delta^-[\Psi](u) = \Delta_{pivot}^-[\Psi](u) \cup \Delta_{rel}^-[\Psi](u)$$

$$\Delta^+[\Psi](u) = \Delta_{pivot}^+[\Psi](u) \cup \Delta_{rel}^+[\Psi](u)$$

where:

- $\Delta_{pivot}^+[\Psi](u)$ denotes the contribution from the pivot-relevant effect of Ψ .
- $\Delta_{rel}^+[\Psi](u)$ denotes the contribution from the relation-relevant effect of Ψ .

The rest of this section briefly explains the major steps of Algorithm 1.

Step 1: Computing the pivot-relevant contribution (Lines 9–18).

If Ψ is pivot-relevant to R , then the tuples in D and I are themselves pivot tuples of Ψ .

In this case, the deletion contribution $\Delta_{pivot}^-[\Psi](u)$ consists of the quads in the pre-update RDF state W_0 generated by Ψ from deleted pivot tuples $d \in D$. These quads become invalid after the update and must be removed from the view.

Symmetrically, the insertion contribution $\Delta_{pivot}^+[\Psi](u)$ consists of the quads generated by applying Ψ to inserted pivot tuples $i \in I$, evaluated over the post-update database state σ_1 . These quads become valid after the update and must be inserted into the view.

Step 2: Computing the relation-relevant contribution (Lines 19–39).

If Ψ is relation-relevant to R , the update affects the evaluation of Ψ indirectly. In this case, tuples in the pivot relation $R^* = pivot(\Psi)$ whose derivations depend on tuples in D or I may have their RDF contributions modified. The computation proceeds in two steps.

Step 2.1: Affected pivot tuples for deletion (Lines 21–33).

Compute A_{Ψ}^- , the sets of pivot tuples in R^* affected by the deleted tuples in D . Two cases must be distinguished:

- **Single occurrence of R in $path(\Psi)$ (Lines 21–23):** the affected pivot tuples can be identified directly from the post-update state σ_1 , without simulating the pre-update state of R .
- **Multiple occurrences of R in $path(\Psi)$ (Lines 24–27):** the computation simulates the pre-update state of R by binding, in Ψ , occurrences of tuple variables associated with R to tuples in the post-update state of R or in D , but not in I (Lines 25 and 26), while binding the occurrences of any other relation S to tuples in the post-update state of S .

Intuitively, the affected sets contain the pivot tuples whose RDF derivations depend on tuples modified by the update.

For each affected pivot tuple $p^* \in A_{\Psi}^-$, define:

- $S_1[\Psi](p^*)$: quads created by Ψ from p^* in the pre-update RDF state W_0 .
- $S_2[\Psi](p^*)$: quads created by Ψ from p^* in the post-update database state σ_1 .

The deletion contribution associated with p^* is then:

$$\Delta_{rel}^-[\Psi](p^*) = S_1[\Psi](p^*) \setminus S_2[\Psi](p^*)$$

Intuitively, the set difference removes from the view exactly the quads that were valid before the update but are no longer generated after the update.

Step 2.2: Affected pivot tuples for insertion (Lines 34–38).

Symmetrically, for each affected pivot tuple $p^* \in A_{\Psi}^+$, the insertion contribution is defined as the set of quads generated by Ψ from p^* in the post-update state σ_1 :

$$\Delta_{rel}^+[\Psi](p^*) = \{(s, p, o, g) \mid (s, p, o) \in \Psi[r^* \leftarrow p^*](\sigma_1), g = URI(\Psi)\}$$

Step 3: Overall deletion and insertion contributions (Lines 40–43).

The overall deletion and insertion contributions of Ψ are then obtained from the combined pivot and relation-relevant contributions:

$$\Delta^-[\Psi](u) = \Delta_{pivot}^-[\Psi](u) \cup \Delta_{rel}^-[\Psi](u)$$

$$\Delta^+[\Psi](u) = \Delta_{pivot}^+[\Psi](u) \cup \Delta_{rel}^+[\Psi](p^*)$$

This concludes the brief remarks on Algorithm 1.

4.3 Correctness and Minimality

This section states a series of results showing that the incremental formulation of changeset computation is correct (see [18] for the proofs). Specifically, the results show that applying the incremental changeset to the pre-update RDF state yields exactly the RDF state of the view after the update. The section also states that the changesets satisfy a minimality criterion.

In what follows, let $\mathcal{W} = (V, \mathcal{S}, M)$ be an RDB2RDF view, $u = (D, I)$ be an update on a relation R , σ_0 and σ_1 be the database states before and after applying u , and $W_1 = RDF_State(\mathcal{W}, \sigma_1)$ and $W_0 = RDF_State(\mathcal{W}, \sigma_0)$ be the view states before and after

Algorithm 1: COMPUTE_DELTA_R(W, u, R, σ_1, W_0)

Input: RDB2RDF view $W = (V, S, M)$; update $u = (D, I)$ over relation R ; post-update database state σ_1 ; pre-update RDF state W_0 .

Output: Changeset $\langle \Delta^-(u), \Delta^+(u) \rangle$.

```
1.  $\Delta^-(u) \leftarrow \emptyset$ ;
2.  $\Delta^+(u) \leftarrow \emptyset$ ;
3.  $Relev(R) \leftarrow \{\Psi \in M \mid \Psi \text{ is pivot-relevant to } R \vee \Psi \text{ is relation-relevant to } R\}$ ;
4. foreach  $\Psi \in Relev(R)$  do
5.    $\Delta_{pivot}^-[\Psi](u) \leftarrow \emptyset$ ;
6.    $\Delta_{rel}^-[\Psi](u) \leftarrow \emptyset$ ;
7.    $\Delta_{pivot}^+[\Psi](u) \leftarrow \emptyset$ ;
8.    $\Delta_{rel}^+[\Psi](u) \leftarrow \emptyset$ ;
   // 1. Pivot-relevant contribution
9.   if  $\Psi$  is pivot-relevant to updates on  $R$  then
10.    foreach  $d \in D$  do
11.       $\Delta_{pivot}^-[\Psi](d) \leftarrow \{(s, p, o, g) \in W_0 \mid s = URI(d) \wedge g = URI(\Psi)\}$ ;
12.       $\Delta_{pivot}^-[\Psi](u) \leftarrow \Delta_{pivot}^-[\Psi](u) \cup \Delta_{pivot}^-[\Psi](d)$ ;
13.    end
14.    foreach  $i \in I$  do
15.       $\Delta_{pivot}^+[\Psi](i) \leftarrow \{(s, p, o, g) \mid (s, p, o) \in \Psi[r \leftarrow i](\sigma_1) \wedge g = URI(\Psi)\}$ ;
16.       $\Delta_{pivot}^+[\Psi](u) \leftarrow \Delta_{pivot}^+[\Psi](u) \cup \Delta_{pivot}^+[\Psi](i)$ ;
17.    end
18.  end
   // 2. Relation-relevant contribution
19.  if  $\Psi$  is relation-relevant to updates on  $R$  then
20.     $R^* \leftarrow pivot(\Psi)$ ;
    // 2.1 Affected pivot tuples for deletion
21.    if  $R$  occurs only once in  $path(\Psi)$  then
22.       $A_{\Psi}^- \leftarrow \{p^* \in R^*(\sigma_1) \mid \exists d \in D, \exists (s, p, o) \in \Psi[r^* \leftarrow p^*, r_R \leftarrow d](\sigma_1)\}$ ;
23.    end
24.    else
25.       $\sigma_R^0 \leftarrow (R(\sigma_1) \cup D) \setminus I$ ;
26.       $A_{\Psi}^- \leftarrow \{p^* \in R^*(\sigma_1) \mid \exists d \in D, \exists r_k \in Var_R(\Psi), \exists (s, p, o) \in \Psi[r^* \leftarrow p^*, r_k \leftarrow d](\sigma_1[R \leftarrow \sigma_R^0])\}$ ;
27.    end
28.    foreach  $p^* \in A_{\Psi}^-$  do
29.       $S_1[\Psi](p^*) \leftarrow \{(s, p, o, g) \in W_0 \mid s = URI(p^*) \wedge g = URI(\Psi)\}$ ;
30.       $S_2[\Psi](p^*) \leftarrow \{(s, p, o, g) \mid (s, p, o) \in \Psi[r^* \leftarrow p^*](\sigma_1) \wedge g = URI(\Psi)\}$ ;
31.       $\Delta_{rel}^-[\Psi](p^*) \leftarrow S_1[\Psi](p^*) \setminus S_2[\Psi](p^*)$ ;
32.       $\Delta_{rel}^-[\Psi](u) \leftarrow \Delta_{rel}^-[\Psi](u) \cup \Delta_{rel}^-[\Psi](p^*)$ ;
33.    end
    // 2.2 Affected pivot tuples for insertion
34.     $A_{\Psi}^+ \leftarrow \{p^* \in R^*(\sigma_1) \mid \exists i \in I, \exists r_k \in Var_R(\Psi), \exists (s, p, o) \in \Psi[r^* \leftarrow p^*, r_k \leftarrow i](\sigma_1)\}$ ;
35.    foreach  $p^* \in A_{\Psi}^+$  do
36.       $\Delta_{rel}^+[\Psi](p^*) \leftarrow \{(s, p, o, g) \mid (s, p, o) \in \Psi[r^* \leftarrow p^*](\sigma_1) \wedge g = URI(\Psi)\}$ ;
37.       $\Delta_{rel}^+[\Psi](u) \leftarrow \Delta_{rel}^+[\Psi](u) \cup \Delta_{rel}^+[\Psi](p^*)$ ;
38.    end
39.  end
   // 3. Overall deletion and insertion contributions
40.   $\Delta^-[\Psi](u) \leftarrow \Delta_{pivot}^-[\Psi](u) \cup \Delta_{rel}^-[\Psi](u)$ ;
41.   $\Delta^+[\Psi](u) \leftarrow \Delta_{pivot}^+[\Psi](u) \cup \Delta_{rel}^+[\Psi](u)$ ;
42.   $\Delta^-(u) \leftarrow \Delta^-(u) \cup \Delta^-[\Psi](u)$ ;
43.   $\Delta^+(u) \leftarrow \Delta^+(u) \cup \Delta^+[\Psi](u)$ ;
44. end
45. return  $\langle \Delta^-(u), \Delta^+(u) \rangle$ ;
```

applying u . Let $\Delta^-(u)$ and $\Delta^+(u)$ be the deletion and insertion components defined as in Section 4.2.

Let σ be a relational database state. Recall that $\mathcal{P}(\mathcal{W})$ denotes the set of pivot relations of $\mathcal{W}$ and the RDF state of $\mathcal{W}$ under σ is given by:

$$RDF_State(\mathcal{W}, \sigma) = \bigcup_{R^* \in \mathcal{P}(\mathcal{W})} \bigcup_{p^* \in R^*(\sigma)} RDF_State[R^*](p^*, \sigma)$$

The first lemma shows that each quad has a unique derivation associated with a single pivot tuple and transformation rule.

LEMMA 4.1 (UNIQUENESS OF QUAD GENERATION). *For every quad $(s, p, o, g) \in RDF_State(\mathcal{W}, \sigma)$, there exists a unique transformation rule $\Psi \in M$ and a unique pivot tuple $p^* \in pivot(\Psi)(\sigma)$ such that*

$$(s, p, o, g) \in RDF_State[\Psi](p^*, \sigma)$$

The next two lemmas show that the incremental deletion and the incremental insertion components capture exactly the quads whose derivations are invalidated by deletions, or created by insertions.

Recall that a quad q has a derivation under state σ iff there is a rule Φ , with pivot relation T^* , and a pivot tuple $t^* \in T^*(\sigma)$ such that $q \in \Phi[r \leftarrow t^*](\sigma)$.

LEMMA 4.2 (CHARACTERIZATION OF THE DELETION COMPONENT).

- (a) (For any quad q) $((q \in W_0 \wedge q \notin \Delta^-(u)) \Rightarrow q \in W_1)$.
- (b) (For any quad q) $(q \in W_1 \Rightarrow (q \notin \Delta^-(u) \vee q \in \Delta^+(u)))$.

LEMMA 4.3 (CHARACTERIZATION OF THE INSERTION COMPONENT). $\Delta^+(u)$ is the set of quads whose derivation becomes valid in σ_1 due to some tuple $i \in I$.

Theorem 4.4 states that the insertion and deletion changesets are correct. The proof follows from the previous lemmas [18].

THEOREM 4.4 (CORRECTNESS OF THE INCREMENTAL CHANGESETS).

$$W_1 = (W_0 \setminus \Delta^-(u)) \cup \Delta^+(u)$$

Finally, Theorem 4.4 has a simple corollary. Let σ_D be the database state after applying the deletions in D , but before applying the insertions in I . Let $W_D = RDF_State(\mathcal{W}, \sigma_D)$ be the view state corresponding to σ_D .

COROLLARY 4.5 (MINIMALITY OF THE INCREMENTAL CHANGESETS).

- (a) $\Delta^-(u) = (W_0 \setminus W_D)$
- (b) $(\Delta^+(u) \setminus W_D) = (W_1 \setminus W_D)$
- (c) If $(\Delta^+(u) \cap W_D) = \emptyset$, then $\Delta^+(u) = (W_1 \setminus W_D)$

Corollary 1 suggests that the changesets are minimal in the following sense. Part (a) says that $\Delta^-(u)$ is exactly the set of quads that are removed from the initial view state W_0 as a result of the deletions in the update u (but before the insertions). Likewise, Part (b) states that $\Delta^+(u)$ is the set of quads that are inserted into the intermediate view state W_D , created after the deletions, as a result

of the insertions in the update u . But we have to ignore quads that are already in W_D , since we do not rule out stale insertions that lead to duplicates. Part (c) is a simple consequence of Part (b) and says that, if $\Delta^+(u)$ has no duplicates w.r.t. W_D , then $\Delta^+(u)$ is exactly the set of quads that must be inserted into W_D to create W_1 .

5 Operationalization of the Formal Model

5.1 Motivation

A natural question arising from the previous sections is whether the formal model can be implemented in practice. To answer this question, this section introduces an LLM-based prototype translation workflow that starts with R2RML mappings that define an entity-preserving RDB2RDF view and produces the required AFTER triggers and related artifacts to maintain the view incrementally.

It should be stressed that the goal of this section is not to introduce a production-ready compiler or a new LLM code-generation method. Instead, the section uses an LLM as an experimental instrument to investigate whether the formal model is sufficiently precise and constructive to guide the generation of executable incremental-maintenance logic. The LLM is not used to establish the correctness of the proposed framework, which follows from the formal results.

This section is structured in two parts. Section 5.2 introduces an asynchronous architecture that re-engineers Algorithm 1 to separate access to the post-update database state σ_1 from access to the pre-update view state W_0 , since the database and the view are typically in distinct environments. Section 5.3 then describes the LLM-based prototype translation workflow that produces the required artifacts to maintain the view incrementally.

5.2 An Asynchronous Architecture

The asynchronous architecture, shown in Figure 2, combines AFTER triggers with an asynchronous maintenance worker.

Instead of directly computing and applying the changeset inside a relational transaction, a trigger captures a snapshot of the update event on a relation R (the transition tables D and I), and computes the affected tuples for deletion $A^-[\Psi](u)$, the post-update contributions $S_2[\Psi](u)$ to $\Delta^-(u)$, and the insertion changeset $\Delta^+(u)$ (which requires computing the affected tuples for insertion $A^+[\Psi](u)$). This information is then stored in a maintenance queue. The AFTER trigger for a relation R calls the function `COMPUTE_CHANGESET_R`, shown in Algorithm 2, to compute $A^-[\Psi](u)$, $S_2[\Psi](u)$, and $\Delta^+[\Psi](u)$ and update the queue.

An external worker processes queued events sequentially, preserving the update commit order. For each event, the worker first queries the triplestore to retrieve from the RDB2RDF view state W_0 the corresponding deletion contribution of deleted tuples, $\Delta_{pivot}^-[\Psi](u)$, and the pre-update RDF contributions, $S_1[\Psi](u)$. Then, it computes the deletion contribution using the set difference $(S_1[\Psi](u) \setminus S_2[\Psi](u))$, applies $\Delta^-(u)$ to the view, and then applies $\Delta^+(u)$.

This architecture offers several advantages. First, it avoids executing SPARQL queries inside relational transactions, preventing triplestore latency or temporary unavailability from blocking database updates. Second, it stores the relational context of the update at commit time, ensuring that subsequent database updates do not affect the computation of the changeset. Third, the queue-based design helps improve recoverability, since maintenance events can

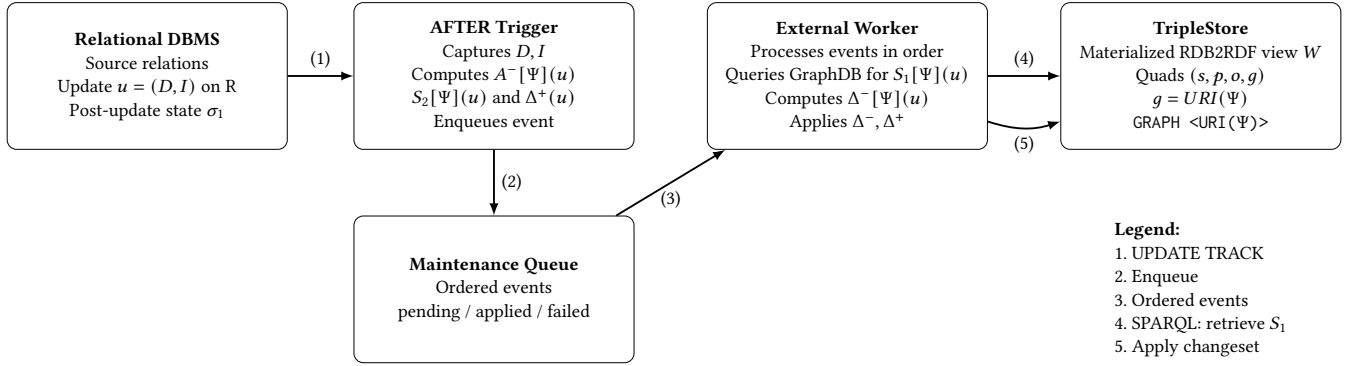


Figure 2: Asynchronous architecture for incremental maintenance of a materialized RDB2RDF view.

Algorithm 2: Function COMPUTE_CHANGESET_R for the Asynchronous Architecture

Input: RDB2RDF view $W = (V, S, M)$; update $u = (D, I)$ over relation R .

Output: Insertions on tables *rdf_maintenance_queue* and *rdf_rule_contribution*.

// Event-level queue initialization:

1. Insert one row into *rdf_maintenance_queue* for the statement-level update $u = (D, I)$;
// Rule-level computation and publication:
 2. **foreach** $\Psi \in \text{Relev}(R)$ **do**
 - // Execute the following two steps sequentially:
 - // STEP 1 – Incremental Maintenance Computation for Ψ
 - 3. $\Delta_{\text{pivot}}^+[\Psi](u) \leftarrow \emptyset$;
 - 4. $\Delta_{\text{rel}}^+[\Psi](u) \leftarrow \emptyset$;
 - 5. **if** Ψ is pivot-relevant to updates on R **then**
 - 6. | Compute $\Delta_{\text{pivot}}^+[\Psi](u)$;
 - 7. **end**
 - 8. **if** Ψ is relation-relevant to updates on R **then**
 - 9. | Compute $A^-[\Psi](u)$, $A^+[\Psi](u)$, $S_2[\Psi](u)$, and $\Delta_{\text{rel}}^+[\Psi](u)$;
 - 10. **end**
 - 11. $\Delta^+[\Psi](u) \leftarrow \Delta_{\text{pivot}}^+[\Psi](u) \cup \Delta_{\text{rel}}^+[\Psi](u)$;
 - // STEP 2 – Rule Contribution Publication
 - 12. Insert one row into *rdf_rule_contribution* for Ψ with $A^-[\Psi](u)$, $S_2[\Psi](u)$, and $\Delta^+[\Psi](u)$
linked to the *event_id* created in *rdf_maintenance_queue*;
 14. **end**
-

be retried or processed asynchronously without impacting transactional workloads.

5.3 An LLM-based Prototype Translation Workflow

The translation workflow implements a two-phase process. Each phase is implemented by a specialized LLM-based module that generates a well-defined intermediate artifact. Since LLMs may occasionally produce incomplete or inconsistent outputs, each generated artifact is independently validated before being accepted as the input to the subsequent phase. This validation process ensures that the output of each phase conforms to the formal model.

Phase 1 – Translating R2RML Statements into Transformation Rules

The first phase translates an Entity-Preserving R2RML mapping into a validated set of transformation rules. An LLM-assisted transformation rules translation module implements this phase, guided by a prompt (available in the supplemental material), whose inputs consist of the relational database schema and the R2RML mapping.

This module performs the following tasks:

- (1) Identifies pivot relations and verifies entity preservation.
- (2) Derives URI constructors and the corresponding hasURI predicates.
- (3) Validates the pivot relations and URI constructors.
- (4) Generates and validates the transformation rules.

The output is a structured JSON specification representing a validated intermediate artifact. R2RML mappings that fail validation are reported together with the detected inconsistencies and

marked as “Manual Analysis Required”. No transformation rules are generated for invalid mappings.

This intermediate representation completely separates semantic reasoning from code generation. Consequently, the second translation phase no longer needs to interpret the R2RML mapping and instead relies exclusively on the validated transformation rules.

Phase 2 – Translating Transformation Rules into AFTER Triggers

The second phase translates the transformation rules generated during Phase 1 into AFTER triggers, as discussed in Section 5.2. An LLM-assisted trigger translation module implements this phase, guided by a prompt (available at the supplemental material), whose inputs consist of the transformation rules generated during Phase 1, and the incremental maintenance framework of Section 5.2.

Each trigger captures a snapshot of the update event on a relation R (the transition tables D and I), and computes the affected tuples for deletion $A^-[\Psi](u)$, the post-update contributions $S2[\Psi](u)$ to $\Delta^-(u)$, and the insertion changeset $\Delta^+(u)$ (which requires computing the affected tuples for insertion $A^+[\Psi](u)$). The resulting code operates on the relational database and communicates with the asynchronous maintenance worker through the queue, without directly accessing the materialized RDF view.

6 Operational Feasibility Study

The goal of this section is to investigate the operationalization of the formal model proposed in Section 5. As a case study, it adopts the LinkedBrainz RDB2RDF View, defined on top of the MusicBrainz relational database using a set of R2RML statements, and validates the complete cycle from R2RML statements to changeset computation.

The goal is specialized into the following questions:

- RQ1 – Correctness of the Transformation Rules Translation:** Can R2RML statements be correctly translated into transformation rules?
- RQ2 – Correctness of the AFTER Triggers Translation:** Can transformation rules be correctly translated into executable AFTER triggers and the maintenance worker?
- RQ3 – End-to-End Correctness:** Do the generated changesets specified by the transformation rules produce the same RDF state as a complete recomputation?
- RQ4 – Human Intervention:** Which parts of the operationalization require human decisions, and why?

The supplemental material contains the complete data used in the case study.

6.1 Setup

Data and LLMs. The experiments were based on:

- *Relational Database:* a local copy of the *MusicBrainz* database, stored in PostgreSQL.
- *LinkedBrainz RDB2RDF View:* an RDB2RDF view of the *MusicBrainz* database, materialized in GraphDB.
- *Selected relations:* the relations *ARTIST*, *MEDIUM*, and *TRACK*, chosen because they occur in complex transformation rules.
- *Selected updates:* one insertion, one deletion, and one update per selected relation, with the expected changesets.

- *Selected LLMs:* GPT-5.6 using the CODEX desktop, and Qwen3.8-Max-Preview in “auto” mode using the free-tier chat interface, referred to simply as GPT-5.6 and Qwen3.8 in what follows.

MusicBrainz⁴ is a large and actively maintained open music knowledge base operated by the MetaBrainz Foundation. The complete relational schema comprises 374 tables, with data about artists, release groups, etc. The LinkedBrainz RDB2RDF View uses the Music Ontology vocabulary and is specified in R2RML through 57 TriplesMaps⁵.

6.2 Correctness Validation Procedure

Transformation Rules Translation Correctness

This step addresses RQ1 and, in part, RQ4. GPT-5.6 was prompted to translate the R2RML statements of the case study into transformation rules. Then, the results were verified, separating URI constructor from transformation rules translation correctness validation. GPT-5.6 also helped in iteratively refining the translation prompt until correct, traceable transformation rules were produced.

URI constructor correctness validation checks the preservation of the complete subject template, placeholder expressions, and ordering, as well as encoding semantics and consistency across mappings that represent the same semantic entity. Transformation rules translation correctness validation checks the RDF class or property, RDF term type, selection conditions, value transformations, null handling, and relational paths.

When the R2RML input contains ambiguities, conflicting URI templates, malformed SQL expressions, unresolved relational paths, or potentially invalid IRI values, the translation phase still generates a candidate rule and marks it for human validation. The output includes the supporting evidence and the issue requiring review. A candidate is validated only after passing all automatic checks or after the corresponding human decision is explicitly recorded.

AFTER Trigger Translation Correctness

This step addresses RQ2 and, in part, RQ4. GPT-5.6 and Qwen3.8 were prompted to translate the transformation rules for the selected relations – *ARTIST*, *MEDIUM*, and *TRACK* – into AFTER Triggers.

For each combination of LLM and selected relation, the correctness of the generated trigger code was manually validated according to the criteria presented in Table 3.

End-to-End Correctness

This step addresses RQ3. The AFTER triggers for the selected relations were executed using different update scenarios. To be more informative, correctness validation was subdivided into two tasks.

Rule Contribution Correctness. Rule contribution correctness validation verifies whether the AFTER triggers compute the expected maintenance data for each relevant transformation rule and update scenario.

⁴Available at <https://musicbrainz.org>

⁵Available at <https://docs.openlinksw.com/virtuoso/rdfviewsbusintmbr/>

In more detail, for each update u and each relevant transformation rule $\Psi \in \text{Relev}(R)$, the generated rule contributions

$$A^-[\Psi], A^+[\Psi], S2[\Psi], \Delta_{pivot}^+[\Psi], \Delta_{rel}^+[\Psi], \Delta^+[\Psi]$$

were manually compared against the expected results derived from the formal incremental maintenance semantics of Section 4.

Changeset Correctness. This step fully addresses RQ3 by providing end-to-end validation of the correctness of the generated maintenance code. It verifies whether the generated changesets produce the same RDF view as a full recomputation, which is independently obtained.

Let W_0 denote the RDF view before the update and $W_1 = M(\sigma_1)$ the RDF view obtained by applying the complete RDB2RDF mapping to the updated database state σ_1 . The generated triggers are considered correct iff the RDF view reconstructed from their computed changesets satisfies $W_1 = (W_0 - \Delta^-(u)) \cup \Delta^+(u)$.

Table 3: Trigger Translation and End-to-End Correctness.

Criteria	Description
<i>AFTER Trigger Translation Correctness</i>	
Transformation Rule Identification	Does the LLM-assisted trigger translation module correctly identify the relevant transformation rules?
Rule Contribution Computation	Does the LLM-assisted trigger translation module correctly generate the code for the rule contributions?
Code Minimality	Does the LLM-assisted trigger translation module generate code which is concise and free of unnecessary operations?
Hallucination	Does the LLM-assisted trigger translation module introduce unsupported assumptions or irrelevant joins?
<i>End-to-End Correctness</i>	
Rule Contribution Correctness	Do the generated triggers compute the expected rule contributions for each relevant transformation rule and update scenario?
Changeset Correctness	Do the generated triggers compute the expected changesets for each update scenario?

6.3 Results

Transformation Rules Translation Correctness

All R2RML statements of the case study were classified as entity-preserving and translated into 89 transformation rules. The correctness of 78 transformation rules was automatically validated, but 11 (12.4%) required human validation, primarily due to malformed SQL joins or mapping inconsistencies.

AFTER Trigger Translation and End-to-End Correctness

Table 4 summarizes the AFTER trigger translation and end-to-end correctness validation for the TRACK relation, with the triggers obtained with GPT-5.6 and Qwen3.8. (see the extended paper [18] for the correctness validation for ARTIST and MEDIUM).

To summarize, the results suggest that:

- GPT-5.6 correctly translated all R2RML statements into transformation rules, but some rules required human validation.
- GPT-5.6 generated correct and optimized AFTER trigger code in most cases.
- Qwen 3.8-Max-Preview generated correct and concise AFTER trigger code in some cases.

Table 4: Trigger Translation and End-to-End Correctness Validation – Results for Relation Track – GPT-5.6 and Qwen3.8.

Criteria	Model	Result
<i>AFTER Trigger Translation Correctness</i>		
Transformation rule Identification	GPT	Identified all relevant rules.
	Qwen	Identified all relevant rules.
Rule Contribution Computation	GPT	Correctly computed all rule contributions.
	Qwen	Correctly computed all rule contributions.
Code Minimality	GPT	Produced a redundant intermediate code.
	Qwen	Produced a redundant intermediate code.
Hallucination	GPT	Did not hallucinate.
	Qwen	Did not hallucinate.
<i>End-to-End Correctness</i>		
Rule Contribution Correctness	GPT	The triggers generated contributions for all 7 rules evaluated. The generated quadruples are correct for 5 of the rules, while the remaining 2 rules show discrepancies regarding literal datatypes.
	Qwen	Of the contributions generated by the 7 rules, 4 produced correct quadruples. A fifth rule was considered correct with the inclusion of <i>xsd:string</i> . The other 2 differ only in the data types of the literals
Changeset Correctness	GPT	The computed changesets correspond exactly to the expected changesets.
	Qwen	(Failed to execute)

7 Conclusions

The first contribution of this paper was a formal model of the deletion and insertion components of RDB2RDF changesets. Existing formulations of incremental RDB2RDF maintenance depend on the pre-update relational state when computing deletion changesets. This creates a mismatch with statement-level AFTER triggers, which execute when the database is already in its post-update state and have direct access only to the transition tables and the current database state. This paper introduces a formal model that removes this mismatch through a post-update formulation and proves the correctness of the resulting changesets.

The second contribution was to describe an LLM-based prototype translation workflow that starts with R2RML statements that define an entity-preserving RDB2RDF view and produces the required AFTER triggers and related artifacts to maintain the view incrementally. The prototype provides evidence that the formal model is sufficiently constructive to support generating executable maintenance triggers from the RDB2RDF view mapping specification.

The third contribution was to conduct an experiment using a real-world database and an RDB2RDF view specified with R2RML statements, illustrating that an LLM, guided by appropriate prompts and accessing the formal model, can generate code to compute and publish the changesets.

As future work, the LLM-based prototype translation workflow will be re-engineered as a multi-agent system with human-in-the-loop to facilitate compiling R2RML statements into AFTER triggers and installing them in the source relational database.

Supplemental Material Statement. The extended paper, with the formal definitions, results, and proofs, and the code, prompts, database, and RDB2RDF view adopted in the case study are available at <https://github.com/semantic-ekgraphs/IVM.git>

Declaration of use of Generative AI. In Sections 4.2 and 5, ChatGPT was used to help check the definitions and define the architecture. In Section 6, GPT-5.6 and Qwen3.8 were utilized to generate code.

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